Bayesian Probit Regression via Latent Variable Gibbs Sampling

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Binary Probit Model: Motivation

- We observe binary outcomes:

$$Y_i \in \{0, 1\}, \quad i = 1, \ldots, N$$

- We model the probability of success using a probit link:

$$Y_i \sim \text{Bernoulli}\big(\Phi(\mathbf{x}_i'\beta)\big)$$

  where $\Phi(\cdot)$ is the standard normal CDF.
- Idea: introduce a **latent Gaussian variable** Z_i to make the posterior tractable and enable Gibbs sampling.

Latent Gaussian Variable Representation

- Introduce a continuous latent variable Z_i:

$$Z_i = \mathbf{x}_i'\beta + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

- Observed binary Y_i is obtained by **thresholding** Z_i:

$$Y_i = \begin{cases} 1 & \text{if } Z_i > 0 \\ 0 & \text{if } Z_i \leq 0 \end{cases}$$

- Intuition:
  - We only see whether Z_i is above or below the cutoff at 0.
  - This construction makes the model conditionally Gaussian in Z and β.

Non-Informative Prior on (β, σ^2)

$$\tau(\beta, \sigma^2) \propto \frac{1}{\sigma^2}$$

- This is a **Jeffrey's-type** non-informative prior.
- Interpretation:
  - Flat (improper) prior on $\beta \Rightarrow$ we do not impose strong beliefs on regression coefficients.
  - Prior on σ^2 is favoring no particular variance level.
- Goal:
  - Let the **data drive** the posterior.
  - Keep the math simple while staying within a Bayesian framework.

Joint Posterior Distribution

The joint posterior for (β, σ^2, Z) is proportional to:

$$p(\beta, \sigma^2, Z \mid y) \propto \prod_{i=1}^{N} [I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0)]$$
$$\exp\left(-\frac{(Z - X\beta)'(Z - X\beta)}{2\sigma^2}\right)(\sigma^2)^{-(N/2+1)}$$

- The indicator term

$$I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0)$$

  enforces that Z_i is **consistent** with the observed Y_i.
- The Gaussian term comes from $Z \mid \beta, \sigma^2 \sim N(X\beta, \sigma^2 I)$.

Fully Conditional Posterior Distributions

- **Variance** σ^2:

$$\sigma^2 \mid \beta, Z, Y \sim \text{Inverse-Gamma}\left(\frac{N}{2}, \frac{(Z - X\beta)'(Z - X\beta)}{2}\right)$$

- **Regression coefficients** β:

$$\beta \mid \sigma^2, Z, Y \sim N\left((X'X)^{-1}X'Z,\ \sigma^2(X'X)^{-1}\right)$$

- **Latent variables** Z_i:

$$Z_i \mid \beta, \sigma^2, Y \sim \text{Normal}\left(\mathbf{x}_i'\beta, \sigma^2\right) \text{ truncated to } \begin{cases} (0, \infty) & \text{if } y_i = 1 \\ (-\infty, 0] & \text{if } y_i = 0 \end{cases}$$

- These three conditionals are all in **standard families**: Inverse-Gamma, Normal, and truncated Normal.
- This makes a **Gibbs sampler** straightforward to implement.

Identifiability Issue in (β, σ^2)

$$P(Y_i = 1 \mid \beta, \sigma^2) = P(Z_i > 0) = \Phi\left(\frac{\mathbf{x}_i'\beta}{\sigma}\right)$$

- The probability depends only on the **ratio**:

$$\theta = \frac{\beta}{\sigma}$$

- As a result:
  - β and σ^2 are **not separately identifiable** from the data.
  - Many combinations of (β, σ^2) produce the same probabilities.

Experimental Data: Insecticide Dose–Response Study

- The study measures how many insects show an **adverse reaction** at different dose levels.
- Each dose level exposes $N_i = 6$ insects.
- A_i denotes the number that reacted adversely.
- Table 5-1 shows the **grouped counts** for each dose.
- To fit a probit model, we must convert grouped data into **individual (dose, response)** pairs.

Table 5-1 Grouped data

Dose d_i	N_i (Exposed)	A_i
1.69	6	1
1.72	6	3
1.76	6	2
1.78	6	3
1.81	6	5
1.84	6	4
1.86	6	6
1.88	6	6

Table 5-2 Individual data

(Dose d_i, Response Y_i) Pairs				
(1.69, 0)	(1.69, 0)	(1.69, 0)	(1.69, 0)	(1.69, 0)
(1.69, 1)	(1.72, 0)	(1.72, 0)	(1.72, 0)	(1.72, 1)
(1.72, 1)	(1.72, 1)	(1.76, 0)	(1.76, 0)	(1.76, 0)
(1.76, 0)	(1.76, 1)	(1.76, 1)	(1.78, 0)	(1.78, 0)
(1.78, 0)	(1.78, 1)	(1.78, 1)	(1.78, 1)	(1.81, 0)
(1.81, 1)	(1.81, 1)	(1.81, 1)	(1.81, 1)	(1.81, 1)
(1.84, 0)	(1.84, 0)	(1.84, 1)	(1.84, 1)	(1.84, 1)
(1.84, 1)	(1.86, 1)	(1.86, 1)	(1.86, 1)	(1.86, 1)
(1.86, 1)	(1.86, 1)	(1.88, 1)	(1.88, 1)	(1.88, 1)
(1.88, 1)	(1.88, 1)	(1.88, 1)	(1.88, 1)	(1.88, 1)

Summary Part - 01

- Built a **Bayesian probit model** using a latent Gaussian variable Z_i.
- Chose a **non-informative prior** $\tau(\beta, \sigma^2) \propto 1/\sigma^2$.
- Derived closed-form **fully conditional distributions** for:
  - σ^2 (Inverse-Gamma), β (Normal), Z_i (truncated Normal).
- Implemented a **Gibbs sampler** to generate posterior draws.
- Identified that only the ratio $\theta = \beta/\sigma$ is **identifiable**.

(Albert & Chib, 1993)

3.3 Hierarchical Analysis

The normal regression structure on $\mathbf{Z}$ also motivates the consideration of normal hierarchical models as presented in Lindley and Smith (1972). Given a particular probit model and regression parameter β of dimension k, one may suspect that β lies on a linear subspace $\mathbf{A}\beta^0$, where β^0 is p-dimensional, where $p < k$. (Alternately, one may believe that the regression parameter satisfies the $(k-p)$-dimensional subspace restriction $\mathbf{\Omega}\beta = 0$. This prior belief may be reexpressed in the above form.) This prior belief can be modeled by the hierarchical model

(1) $\mathbf{Z}$ is distributed $N(\mathbf{X}\beta, \mathbf{I})$,
(2) β is distributed $N(\mathbf{A}\beta^0, \sigma^2\mathbf{I})$, and
(3) (β^0, σ^2) is distributed according to the prior density $\pi(\beta^0, \sigma^2)$. (12)

Hierarchical Analysis

Informative structure:

$$\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2 \sim N(A\boldsymbol{\beta}_0, \sigma^2 I_2)$$

$$\tau(\boldsymbol{\beta}_0) \propto 1, \quad \tau(\sigma^2) \propto \frac{1}{\sigma^2} \text{ (Jeffreys prior)}$$

$$\underbrace{Z \to \boldsymbol{\beta} \to \boldsymbol{\beta}_0 \to \sigma^2 \to Z \to \cdots}_{\text{Hierarchical Gibbs sampler}}$$

Joint Posterior Distribution

The joint posterior for (β, σ^2, Z) is proportional to:

$$\begin{aligned}
\tau(\boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2, z \mid y) &\propto f(y \mid Z, \boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2)\, \tau(Z, \boldsymbol{\beta}, \boldsymbol{\beta}_0, \sigma^2) \\
&\propto \prod_{i=1}^{N} \left[I(Z_i > 0, y_i = 1) + I(Z_i \leq 0, y_i = 0) \right] \\
&\quad \exp\left[-\frac{1}{2} \sum_{i=1}^{N} (Z_i - \mathbf{x}_i' \boldsymbol{\beta})^2 - \frac{1}{2\sigma^2} (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)' (\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0) \right] (\sigma^2)^{-(2/2+1)}
\end{aligned}$$

Fully Conditional Posterior Distributions

- **Hyperparameter** σ^2:

$$\sigma^2 \mid \boldsymbol{\beta}, \boldsymbol{\beta}_0, y \sim IG\left(1,\ \frac{(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)'(\boldsymbol{\beta} - \mathbf{A}\boldsymbol{\beta}_0)}{2}\right) \quad (\sigma^2 \text{ independent of } Z)$$

- **Coefficient** $\boldsymbol{\beta}$:

$$\boldsymbol{\beta} \mid \boldsymbol{\beta}_0, \sigma^2, Z, y \sim N\big((\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2)^{-1}(\mathbf{X}'\mathbf{Z} + \sigma^{-2}\mathbf{A}\boldsymbol{\beta}_0),\ (\mathbf{X}'\mathbf{X} + \sigma^{-2}\mathbf{I}_2)^{-1}\big)$$

- **Latent Variable** Z_i:

$$Z_i \mid \boldsymbol{\beta}, y \sim N(\mathbf{x}_i'\boldsymbol{\beta}, 1) \text{ truncated to } \begin{cases} (0, \infty), & \text{if } y_i = 1, \\ (-\infty, 0], & \text{if } y_i = 0. \end{cases}$$

- **Hyperparameter** $\boldsymbol{\beta}_0$:

$$\boldsymbol{\beta}_0 \mid \boldsymbol{\beta}, \sigma^2 \sim N\big((\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\boldsymbol{\beta},\ \sigma^2(\mathbf{A}'\mathbf{A})^{-1}\big)$$

Posterior Processing: Burn-in & Thinning

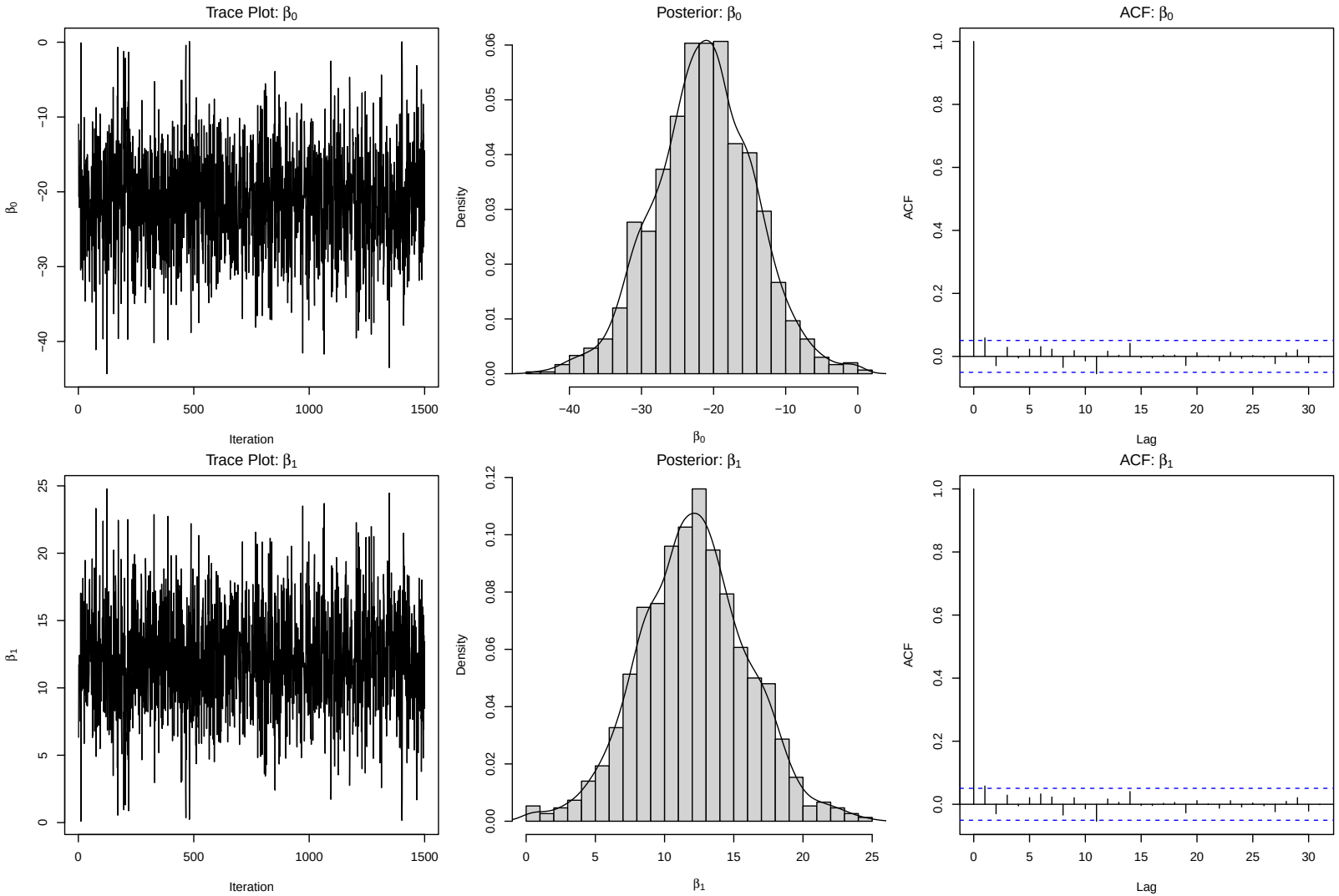
- Early samples discarded — chain has not converged
- **Burn-in:** first 5,000 iterations removed
- **Thinning:** keep every 10th sample

$$\text{Final retained samples} = \frac{20000 - 5000}{10} = 1500$$

  - Reduces autocorrelation
  - Improves independence of stored draws

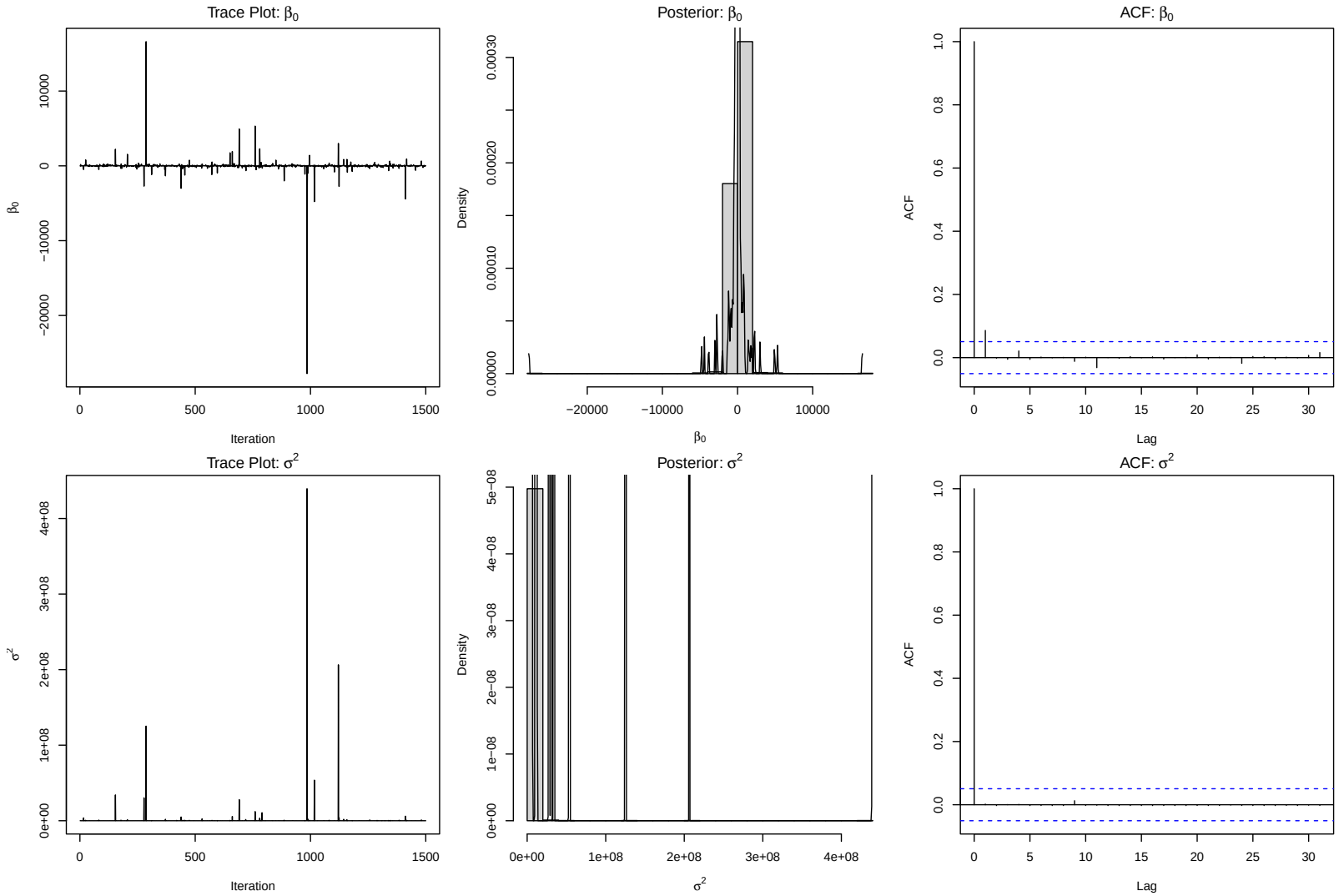
Diagnostics: Coefficients

- β_0, β_1 show stable trace behavior
- Posterior distributions are concentrated and interpretable
- Increasing dose $\Rightarrow$ increasing probability of adverse event



Diagnostics: Hyperparameters

- β_0 and σ^2 posteriors remain diffuse
- Trace plots show wide fluctuations
- Still limited information about latent structure scale



Summary Part - 02

- Latent variable probit model successfully implemented
- Reliable inference on dose-response relationship
- Hyperparameters still weakly informed

References

- Albert, J. H., & Chib, S. (1993). *Bayesian analysis of binary and polychotomous response data*. JASA, 88(422), 669–679.

Thank You!